

code is 192.168.1.1. So assign an IP address to the PC's Ethernet connection, say, 192.168.1.10, with subnet mask 255.255.255.0 (refer Fig. 3). After connecting Ethernet shield to the computer with Ethernet cable, open the web browser (for example, Google Chrome) and type the address as <http://192.168.1.1>.

By default, this will load gauge.htm file from the SD card. To load other files or webpages, menu bars or links are given in the default webpage. It is possible to seamlessly switch between displays with hyperlinks given in each page. If files are not loaded, open the serial monitor of Arduino IDE to check any error message and take appropriate action.

Getting data from Arduino

After the webpage canvas code is loaded from the SD card to the system, input data is obtained from the Arduino. Here the technique used is AJAX (asynchronous JavaScript and XML), which helps to update only the diagram/chart without reloading the entire webpage. AJAX enables

the browser to send and receive data from the server in the background. The webpage sends AJAX requests to the server at regular intervals. The Arduino server responds with an XML (Extensible Markup Language) file containing data values.

XML is much like HTML. It is designed to describe data with tags organised in a tree structure. The tree structure contains root (parent) elements and child elements as defined by the user. Here the XML file is created by Arduino code with `<inputs>` as the root element and `<analog>` and `<digital>` as child elements for analogue and digital inputs, respectively. The XML file created by Arduino is given below:

```
<?xml version="1.0" ?>
<inputs>
  <analog>125 </analog>
  <analog>257 </analog>
  <analog>355 </analog>
  <analog>850 </analog>
  <analog>900 </analog>
  <digital>0 </digital>
  <digital>1 </digital>
  <digital>0 </digital>
  <digital>0 </digital>
</inputs>
```

where the first line is XML declaration.

Displaying graphical data on webpage

After the XML file is received from Arduino, the HTML5 file parses contents with a JavaScript code, which extracts the data and sends it to the canvas display code. The canvas code shows data in the appropriate diagram/chart.

Four types of displays are designed with canvas code as explained below.

Gauge meter. A gauge meter consists of a circular graduated dial and a pointer actuated by a measured input value. Here, the gauge meter is designed by plotting basic circles and lines (refer Fig. 4) in the HTML5 canvas with a few lines of code (gauge.html). The needle is rotated from left to right when the input value is increased.

For demonstration, analogue data input values are taken directly from Arduino's analogue pins A1 through A3. In an actual application, these can be processed values taken from temperature, pressure or humidity sensors with corresponding modifications in the code and dial markings.

Analogue-to-digital converter

(ADC) in the Arduino maps input voltages between 0 and 5 volts as analogue inputs into integer values between 0 and 1023. Hence the gauge meter varies from 0 to 1023.

Bar graph. A bar graph uses horizontal or vertical bars to show the amount of data being measured. Here a simple vertical bar graph with vertical colour

gradient is coded with canvas elements (bargraph.html). You can make any type of graph, such as pie chart and area chart, depending on your requirement.

Any type of data generated by Arduino can be used to plot the graph. As in the gauge meter, here also analogue input values are directly taken as data values to show the working. The six bars correspond to analogue

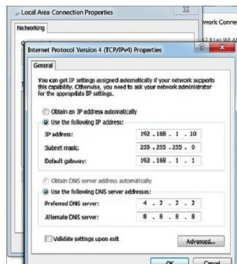


Fig. 3: Setting IP address of the PC's connection

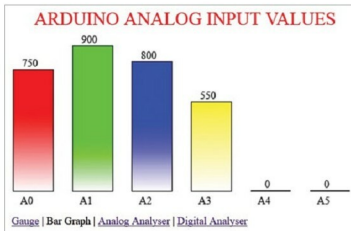


Fig. 5: Bar graph



Fig. 4: Gauge meter readings